Patrick Thangarajah

Simulation & FEA | Testing & Validation | Mechanical Design | Prototyping
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Summary

Recent MASc graduate in mechanical engineering with experience in vehicle safety, impact biomechanics, finite element modelling, physical test data analysis, literature review, and technical writing. Graduate research focused on validating a THOR-5F FE model and evaluating how seat-belt anchor locations affected rear-seat occupant response in frontal impacts.

Skills

Technical Communication	Scientific literature reviews, technical reports, technical presentations, work instructions, SOPs
Design & Prototyping	SolidWorks, Onshape, Engineering Drawings, 3D Printing, Lathe, Mill, Bandsaw, Drill Press
Simulation & Programming	FEA, LS-DYNA, LS-PrePost, Python, C/C++
Controls & Instrumentation	Arduino, Sensors, Motors, Data Acquisition, Circuit Design, Altium, LTspice

Education

Master of Applied Science in Mechanical and Mechatronics Engineering — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*
- Research in impact mechanics, finite element analysis, and model validation

Bachelor of Applied Science in Mechanical Engineering, Co-op Program — *Vancouver, BC, Canada*
University of British Columbia — *Sep 2018 - May 2023*
- Biomechanics and Medical Devices Option

Work/Research Experience

Graduate Research Student - Impact Mechanics and Material Characterization Group — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*
- Designed and analyzed a 104-simulation DOE-based FE parametric study evaluating the effect of seat belt restraint anchor locations on rear seat occupant response, identifying a rearward and inboard position that reduced peak chest deflection by 16% below the study average
- Validated a THOR-5F ATD finite element model against physical sled testing in two bench configurations in LS-DYNA, achieving an average ISO 18571 score of 0.71 across 10 measurement channels
- Investigated material response, contact interactions, boundary conditions, and model behavior to understand discrepancies between simulation predictions and experimental measurements
- Communicated findings through a master's thesis, conference paper, technical reports, and six sponsor presentations

R&D Engineer (Part-time) — *Remote*
Echoscope — *May 2023 - Sep 2023*
- Designed and iterated the bladder testing platform PCB, integrating pressure sensing, Arduino connections, power distribution, and motor and valve interfaces; refined component placement and routing and completed design-rule checks
- Authored five manufacturing and verification procedures covering PCB assembly, wiring, software setup, and functionality testing; all five were approved and used by the client

Mechanical Engineering Intern — *Airdrie, AB, Canada*
GN Corporation — *Jan 2022 - Apr 2022*
- Created standardized work instructions and SOPs verified against engineering drawings and customer specifications

Engineering Intern — *Victoria, BC, Canada*
Breathe Global — *Jul 2021 - Aug 2021*
- Redesigned the analog control circuit for a negative-pressure breathing device in Altium, using LTspice simulation to select low-cost components
- Expanded the device's achievable breathing range from 70% to 88% of target

Projects

Simulation of Cervical Fractures in American Football (ME 720 Course Project) — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Dec 2023*
- Simulated head-first helmet impacts across three velocities (3.4–7.0 m/s) and three neck angles at the lowest speed condition using Hybrid III and GHBMC human body models in LS-DYNA, then evaluated fracture force thresholds and injury location against peer-reviewed literature
- Discovered that predicted peak neck forces (2.26–2.34 kN) fell within the range reported across two cadaveric studies, and that the Neck Injury Criterion exceeded its injury-risk threshold in only 1 of 5 configurations despite fracture occurring in all 5

Soft Distensible Bladder Model - Capstone Project — *Vancouver, BC, Canada*
University of British Columbia — *Sep 2022 - Apr 2023*
- Prototyped bladder phantoms using silicone and agar, comparing pressure-volume response with literature to guide material selection

Conference Presentations and Publications

- **Thangarajah, P.** (2026). *Assessment of seat belt anchor position on rear seat occupant kinematics in frontal impact using a validated THOR-5F finite element model*. Master of Applied Science thesis, University of Waterloo, Waterloo, ON, Canada. hdl.handle.net/10012/23684
- Watson, B., Seeburrun, T., **Thangarajah, P.**, Tang, K., Tylko, S., & Cronin, D. S. (2026). *Experimental and numerical investigation of belt loading for small-stature female occupants in rear seat frontal crash*. Proceedings of the 28th International Technical Conference on the Enhanced Safety of Vehicles (ESV), Toronto, ON, Canada, May 12–15, 2026.
- **Thangarajah, P.**, Watson, B., Tang, K., Tylko, S., & Cronin, D. (2025). *Analysis of rear seat shoulder belt anchor position on kinematics of small-stature female occupants in frontal impact*. Injury Biomechanics Symposium, The Ohio State University, Columbus, OH, USA, May 18–21, 2025.